

Rishabh Dholakia

rd625@cornell.edu | [linkedin.com/in/rishabh-dholakia/](https://www.linkedin.com/in/rishabh-dholakia/) | [Portfolio](#)

Education

Cornell University, College of Engineering
Bachelor of Science in Mechanical Engineering

Ithaca NY
Anticipated May 2028

GPA: 4.185

Relevant Coursework: Intro to Mechanical Design, Engineering Simulations/Designs, Dynamics, Thermodynamics, Statics and Mechanics of Solids

Experience

Engineered Living Materials Institute

Ithaca NY

Undergraduate Research Member

August 2025 - Present

- Design and manufacture micro-robot for novel chemical-based control system for multirobot coordination for future search-and-rescue capabilities
- Co-Developed complete CAD assembly using Fusion 360 for V0 chassis and internal components, optimized for 3D printing and rapid assembly
- Developed an initial run & tumble algorithm to validate component integration and pathfinding logic, ensuring reliable robot navigation before hardware finalization

Liquid Propulsion at Cornell (LPC)

Ithaca NY

Mechanical Engineering Member - Turbopump Subteam Lead

August 2025 - Present

- Organize and delegate tasks over a 3-person subteam to develop a liquid-propellant turbopump for engine system integration
- Design turbine to withstand operating conditions of 25000 RPM and blades for required power output
- Validate component structural integrity and fluid power efficiency using FEA and CFD simulations

Cornell Assistive Technologies

Ithaca NY

Mechanical Engineering Subteam Member

November 2024 - Present

- Engineered a tactile coding system by adapting Microsoft MakeCode syntax into a library of 3D-printed blocks with embedded Braille, enabling visually impaired students to learn programming
- CAD modeled a compact enclosure for a sensory overload watch in Fusion 360, optimizing for electronics protection, user comfort, and wearability
- Modified ride-on toy cars using soldering and hand tools, improving accessibility for 5+ children

ASTRA Lab

Ithaca NY

Undergraduate Research Assistant

January 2025 - May 2025

- Analyzed electrospray thruster data with MATLAB & custom Python code for improved efficiency
- Built interactive Python GUI to streamline experimental data analysis & aid usability for researchers
- Modeled and prototyped mounting plate and stand in Fusion 360 for water jetting

Projects

3D Printed Payload Drone

May 2025 - August 2025

- Designed & manufactured a 3D-printed drone frame with a custom 1lb payload bay using DFM/DFA
- Performed static FEA on frame to validate structural integrity against mission load requirements
- Integrated entire electronic wiring (Brushless Motors, AIO FC, RX, VTX, Battery) through PCB soldering and verified continuity & isolation of components

BotBuddy - Remote-Controlled Mobile Robot

May 2025 - August 2025

- Designed & assembled Raspberry Pi-powered robot with live video streaming and remote control
- Prototyped circuitry on breadboards to integrate DC gear motors and drivers with Raspberry Pi GPIO
- Created Flask-based web interface controlled through Python backend to remotely control the robot

Skills

Fusion 360, FEA, DFM/DFA, MATLAB, Python, VBA/Excel, 3D Printing, Mill/Lathe, Soldering